

# Oluwatobi Melvyn Mayungbo

Machine Learning Engineer

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## SUMMARY

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Machine learning engineer who owns models from problem definition to monitoring. Built sequence models served at 5.1 ms p99, a causal targeting study that decides who to contact under a budget, and a self-operating forecasting pipeline with drift monitoring and retraining. B.Sc. in Statistics.

## SKILLS

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**Machine Learning:** PyTorch, LightGBM, XGBoost, Feature Engineering, Sequence Models (GRU, TCN, Transformers), Scikit-learn, Deep Learning, Model Calibration (Isotonic Regression, Temperature Scaling), SHAP

**Statistics and Causal Inference:** Uplift Modeling, Causal Inference (S/T/X-Learners, Causal Forests), Experiment Design, Statistical Inference, Power Analysis, CUPED, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

**Engineering and MLOps:** Python, SQL, FastAPI, Docker, MLflow, Evidently, Prefect, DuckDB, PostgreSQL, Redis, GCP Cloud Run, AWS ECS Fargate, Railway, dbt-core, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

**AI and LLM Systems:** Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

**Data Analysis:** Pandas, NumPy, SQL, Anomaly Detection

**Payments and Risk:** Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

## PROJECTS

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### Uplift Modelling for Campaign Targeting | [GitHub](#)

*Tech: Python, scikit-learn, LightGBM, EconML, pytest, GitHub Actions*

- Ran a causal inference study on the Hillstrom randomised e-mail trial (64,000 customers) to decide which customers to contact under a fixed budget, delivered as a two-page decision memo for a marketing director.
- Showed the campaign works (1.25% conversion against 0.57%) but that a non-randomised list overstates the effect by 7.5×, and that a response model mails five times as many customers the campaign harms as an uplift model does.
- Evaluated with Qini curves, AUUC, decile tables and bootstrap intervals; uplift targeting was worth about \$1,390 against about \$3,170 for raising the budget, and a failed placebo test on the primary campaign was reported openly.

### Sequence-Based Transaction Fraud Detection System | [GitHub](#)

*Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP*

- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.
- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.

### Self-Operating Demand Forecasting Platform | [GitHub](#)

*Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions*

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.

### Telco Customer Churn, End to End on AWS | [GitHub](#)

*Tech: Python, XGBoost, MLflow, FastAPI, Gradio, Docker, GitHub Actions, AWS ECS Fargate, Application Load Balancer, CloudWatch*

- Served it through a FastAPI /predict endpoint with a Gradio UI, containerised with Docker and deployed on AWS ECS Fargate behind an Application Load Balancer, with CloudWatch logs and GitHub Actions CI/CD.

## EXPERIENCE

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### Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

### Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

## EDUCATION

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### B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

### Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

## CERTIFICATIONS

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- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action